TABLE III

SUPPLEMENTARY TABLE OF CANDIDATE AGENTIC SECURITY SYSTEMS EVALUATED AGAINST ELIGIBILITY CRITERIA (C1, C2, C3)

#	System	Authors	Venue / Identifier	C1	C2	C3	Status	Screening note
Pentesting & Offensive Validation								
1	PentestGPT [28]	Deng et al.	USENIX Sec. 2024	Y	Y	Y	Evaluated	Pentest task tree & text log, L1,L2
2	HackSynth [29]	Muzsai et al.	arXiv:2412.01778 (2024)	Y	Y	Y	Evaluated	Planner, summarizer, cmd/obs log
3	VulnBot [30]	Kong et al.	arXiv:2501.13411 (2025)	Y	Y	Y	Evaluated	Task graph & per-agent logs
4	Incalmo [42]	Singer et al.	IEEE S&P 2026	Y	Y	Y	Evaluated	Abstraction layer, constrained action log
5	PentestAgent [57]	Shen et al.	AsiaCCS 2025 / arXiv:2411.05185	Y	Y	Y	Candidate	RAG multi-agent pentest, L1, L2
6	BreachSeek [58]	Alshehri et al.	arXiv:2409.03789 (2024)	Y	Y	Y	Candidate	Multi-agent auto pentester, L1, L2
7	AutoAttacker [59]	Xu et al.	arXiv:2403.01038 (2024)	Y	Y	N	Excl. (C3)	LLM-guided attack, no analyst-facing output, post-hoc logs only, L1, L2
8	AutoPT [60]	Wu et al.	arXiv:2411.01236 (2024)	Y	Y	Y	Candidate	Web pentest, L1, L2
9	RapidPen [61]	Nakatani	arXiv:2502.16730 (2025)	Y	Y	Y	Candidate	IP-to-shell autonomous, L1, L2
10	PenHeal [62]	Huang & Zhu	Wkshp Auto. Cybersec. 2023	Y	Y	Y	Candidate	Two-stage pentest & remediation, L1, L2
11	RefPentester [63]	Dai et al.	IEEE PST 2025	Y	Y	Y	Candidate	Self-reflective, reflection algo, eval, L1, L2
12	RedTeamLLM [64]	Wang et al.	IFIP AI4KM 2025	Y	Y	Y	Candidate	Agentic offensive framework, L1, L2
13	CIPHER [65]	Pratama et al.	Sensors (2024)	N	Y	Y	Excl. (C1)	Pentest assistant, lacks autonomous execution loop
14	EPPTA [66]	Li et al.	Engineering Reports (2025)	N	Y	N	Excl. (C1)	DRL/POMDP agent, not LLM planning
15	Checkmate [67]	Wang et al.	arXiv:2512.11143 (2025)	Y	Y	Y	Candidate	LLM & classical planning, L1, L2
16	AutoPentest [68]	Henke	arXiv:2505.10321 (2025)	Y	Y	Y	Candidate	Autonomous LLM pentest, L1, L2
17	Shell or Nothing [69]	Mai et al.	arXiv:2509.09207 (2025)	Y	Y	Y	Candidate	Memory-activated multi-agent (Termiagent), L1, L2
18	PentAGI [70]	VXControl (OSS)	Tech. doc. / repo (2026)	Y	N	Y	Excl. (C2)	Open source framework, no documentation found
19	Strix [71]	usestrix (OSS)	Tech. doc. / repo (2026)	Y	N	Y	Excl. (C2)	Open source CLI agent, no documentation found
Incident Response & Detection								
20	IRCopilot [31]	Lin et al.	arXiv:2505.20945 (2025)	Y	Y	Y	Evaluated	Planner/Analyst/Generator/Reflector; L1,L2
21	AutoBnB-RAG [32]	Liu & Anwar	IEEE ICDMW 2025 / arXiv:2508.13118	Y	Y	Y	Evaluated	Multi-agent + RAG; discussion transcripts
22	Audit-LLM [72]	Song et al.	arXiv:2408.08902 (2024)	Y	Y	Y	Candidate	Multi-agent insider-threat detection, L1, L2
23	CORTEX [50]	Wei et al.	arXiv:2510.00311 (2025)	Y	Y	Y	Evaluated	Collaborative agents, high-stakes alert triage, L1, L2
24	Cyber-Sleuth [73]	Fumero et al.	Pre-print (2025)	Y	Y	Y	Candidate	Autonomous blue-team forensics agent, L1, L2
25	MyAntFarm.ai [74]	Drammeh	arXiv:2511.15755 (2025)	Y	Y	Y	Candidate	Multi-agent IR decision support, L1, L2
26	MALCDF [75]	Bharadwaj et al.	arXiv:2512.14846 (2025)	Y	Y	Y	Candidate	Distributed multi-agent cyber defense (MITRE mappings), L1, L2
27	SOCpilot [76]	Barbieri et al.	arXiv:2605.05501 (2026)	Y	Y	Y	Candidate	Multi-step SOC investigation, L1, L2
28	Threat Hunting (Splunk) [77]	Sahay et al.	arXiv:2603.23966 (2026)	Y	Y	Y	Candidate	Agentic threat hunting (SPL plan traces), L1, L2
29	LLMLogAnalyzer [78]	Cai et al.	arXiv:2510.24031 (2025)	N	Y	Y	Excl. (C1)	Clustering-based log QA chatbot
30	IDS-Agent [44]	Li et al.	NeurIPS 2024 Wksp	Y	Y	Y	Evaluated	Explains every detection, L1, L2
31	ProvSEEK [45]	Mukherjee & Kantarcioglu	arXiv:2508.21323 (2025)	Y	Y	Y	Evaluated	Verification-first, analyst forensic summaries, L1, L2
32	OMNISEC [46]	Cheng et al.	arXiv:2503.03108 (2025)	Y	Y	Y	Evaluated	RAG provenance IDS, reconstructs attack graph, L1, L2
33	Anomaly-Agent [12]	Saheed & Chukwuere	IEEE IoT Journal (2025)	Y	Y	Y	Evaluated	Agentic IoT anomaly detection, SHAP, L1, L2, L3
34	Sec-Logs-to-ATT&CK [48]	Hans et al.	arXiv:2510.20930 (2025)	Y	Y	Y	Evaluated	RAG-LLM telemetry segmentation maps IDS logs to ATT&CK + attacker traits, L1, L2
35	Argos [79]	Gu et al.	arXiv:2501.14170 (2025)	Y	Y	Y	Candidate	Agentic time-series anomaly detection, L1, L2
36	POLAR [49]	Tang et al.	arXiv:2510.01552 (2025)	Y	Y	Y	Evaluated	4-stage threat prioritization, L1, L2
37	Graph based multi-agent [47]	Chou et al.	J. Sensor Actuator (2026)	Y	Y	Y	Evaluated	Graph multi-agent (3-phase), SHAP + grounded incident report, L1, L2, L3
Phishing & Threat Analysis								
38	MultiPhishGuard [43]	Xue et al.	arXiv:2505.23803 (2025)	Y	Y	Y	Evaluated	Cooperative agents + PPO, verdict, conf.,rationale, L2
39	PhishDebate [51]	Li et al.	IEEE BigData 2025 / arXiv:2506.15656	Y	Y	Y	Evaluated	Multi-agent debate on phishing sites, L1, L2
40	Debate-Driven MA-LLM [80]	Nguyen et al.	ISDFS 2025 / arXiv:2503.22038	Y	Y	Y	Candidate	Debate, judge agent, L1, L2
41	Multi-Agent Spam [81]	Huang	IEEE ICIBA 2024	Y	Y	Y	Candidate	Spam debate framework, L1, L2
42	Phishing ID LLM [82]	Lee	arXiv:2502.04759 (2025)	N	Y	N	Excl. (C1)	Single-shot LLM classifier
Deployed SOC Copilots & Platforms								
43	MS Security Copilot [52]	Microsoft	Product documentation	Y	Y	Y	Evaluated	Promptbooks + process log; L1,L2
44	Charlotte AI [53]	CrowdStrike	Technical documentation	Y	Y	Y	Evaluated	Bounded autonomy; confidence+priority score
45	D3 Morpheus [54]	D3 Security	Product documentation	Y	Y	Y	Evaluated	Single reasoning engine SOAR; EAR risk score
46	Dropzone AI [55]	Dropzone AI	Technical documentation	Y	Y	Y	Evaluated	Evidence-cited reports; audit trail
47	GreyMatter [56]	ReliaQuest	Product documentation	Y	Y	Y	Evaluated	Six personas; confidence scoring + audit
48	COMET [83]	Wang et al.	IEEE ISSRE 2024	Y	Y	Y	Candidate	Deployed MS triage, interpretable, L1, L2
49	HAIC-SOC [84]	Mohsin et al.	arXiv:2505.23397 (2025)	Y	N	N	Excl. (C2)	Conceptual design framework, n.d.
50	Cognitive SOCs SLR [85]	Binbeshr et al.	IEEE OJ-CS 2025	N	Y	N	Excl. (C1)	Literature review, not a system
51	Cloud Reliability AI [86]	Chigurupati et al.	IEEE ICAIC 2025	N	Y	Y	Excl. (C1)	SRE/cloud availability not in scope
Excluded — Evaluation Benchmarks (C1)								
52	PACEbench [33]	Liu et al.	arXiv:2510.11688 (2025)	N	Y	N	Excl. (C1)	Benchmark; assesses models, not agent
53	CyberSOCEval [34]	Deason et al.	arXiv:2509.20166 (2025)	N	Y	N	Excl. (C1)	Benchmark suite (malware/CTI reasoning)
54	AutoPenBench [35]	Gioacchini et al.	arXiv:2410.03225 (2024)	N	Y	N	Excl. (C1)	Benchmark for generative pentest agents
55	AttackSeqBench [36]	Yong et al.	arXiv:2503.03170 (2025)	N	Y	N	Excl. (C1)	Benchmark of attack-sequence understanding
56	Cybench [37]	Zhang et al.	ICLR 2025	N	Y	N	Excl. (C1)	Capability/risk evaluation framework